

4-bit Processor Design (cpu4)

Custom 16-instruction ISA, multi-cycle FSM control, Python assembler and golden-model verification

AT A GLANCE

16 opcodes, 8-bit words

ISA
3 / 3 PASS (golden match)

Programs verified

3 cycles / instruction

CPI
317 LC, Fmax 115.86 MHz

iCE40 HX8K (nextpnr)

222 LUT4, 124 FF

Yosys synth_ice40

6205 um², 826 cells

CMOS liberty area

2.35 ns, 18 levels

mini_sta critical path

97 LUT, 60 FDRE

Xilinx xc7 (stat)

VERDICT

SIM PASS GOLDEN MATCH

SYNTH OK TIMING MET

UNITTEST 4/4

VIVADO NOT RUN

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Objective and brief mapping

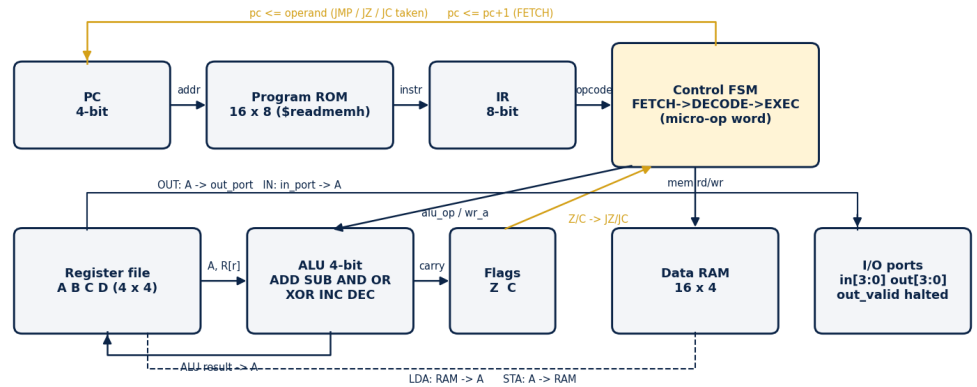
The brief asked for a Verilog 4-bit CPU with a custom 8-bit-instruction ISA, a 4 x 4-bit register file, an ALU with Z/C flags, a multi-cycle fetch/decode/execute control FSM with a documented micro-op table, 16 x 8 program ROM, 16 x 4 data RAM, a Python assembler, three programs verified against a Python ISA simulator, waveform PNGs, iCE40 synthesis with real Fmax, CMOS-liberty area, and Vivado scripts. I implemented every item; the only thing not executed here is Vivado (declared below).

Architecture

Instruction = opcode[7:4] | operand[3:0]. Registers R0 = A (accumulator), R1..R3 = B, C, D. The operand is an immediate for LDI, a RAM address for LDA/STA, a ROM address for JMP/JZ/JC, a register index for ALU ops, and {dst, src} for MOV. Opcode F is a small group: HLT, INC, DEC.

State	Micro-operations performed
FETCH	IR <= ROM[PC]; PC <= PC + 1
DECODE	control word <= f(opcode, operand, Z, C): uop_alu, uop_wr_a, uop_wr_reg, uop_mem_rd, uop_mem_wr, uop_jump (1 / Z / C), uop_out, uop_in, uop_halt, uop_set_c; alu_op in {add, sub, and, or, xor, inc, dec, pass}
EXECUTE	A <= ALU (Z, C updated) A <= RAM[opnd] RAM[opnd] <= A R[dst] <= R[src] A <= in_port out_port <= A + out_valid PC <= opnd HALT

cpu4 -- 4-bit multi-cycle processor datapath and control



Instruction set

Op	Mnemonic	Operand	Action	Flags
0 / 1	NOP / LDI imm	- / immediate	- / A <= imm	- / Z
2 / 3	LDA / STA addr	RAM address	A <= RAM[a] / RAM[a] <= A	-
4	MOV d,s	{dst[1:0], src[1:0]}	R[d] <= R[s]	-
5 / 6	ADD / SUB r	register 0..3	A <= A + R[r] / A - R[r]	Z, C (borrow)
7 / 8 / 9	AND / OR / XOR r	register	A <= A op R[r]	Z
A / B / C	JMP / JZ / JC addr	ROM address	PC <= addr (always / if Z / if C)	-
D / E	IN / OUT	-	A <= in_port / out_port <= A, out_valid	-
F	HLT / INC / DEC	0 / 1 / 2	halt / A <= A + 1 / A - 1	Z, C

Implementation details

- rtl/cpu4.v (145 lines): synchronous reset, 3-state FSM, registered control word, 5-bit ALU result so bit 4 gives carry (ADD/INC) or borrow (SUB/DEC), ROM via \$readmemh parameter so the same RTL runs any program.
- sw/asm.py: two-pass assembler (labels, decimal/hex, register names, MOV d,s packing, 16-word limit) writing .hex and a .lst listing. sw/isa_sim.py: golden model that emits the OUT stream and expected.hex.
- Programs: prog1_add (STA/LDA/ADD/IN, outputs C then A), prog2_countdown (9..1 via SUB/JZ, then 0), prog3_fib (1 1 2 3 5 8 13, exits on JC when 13+8 overflows 4 bits).
- Testbench tb/tb_cpu4.v reads expected.hex, compares every out_valid pulse, counts cycles and \$fatal-s on mismatch, extra output or timeout; tb/test_sw.py holds 4 unittest cases for the software.

Verification (Icarus Verilog 14)

Program	Golden OUT sequence	RTL	Instr.	Cycles	Result
prog1_add	C, A	identical	13	39	PASS
prog2_countdown	9 8 7 6 5 4 3 2 1 0	identical	40	120	PASS
prog3_fib	1 1 2 3 5 8 D	identical	44	132	PASS
tb/test_sw.py	asm encoding, 16-word limit, fib golden, SUB borrow/Z/JC	-	-	-	4/4 OK

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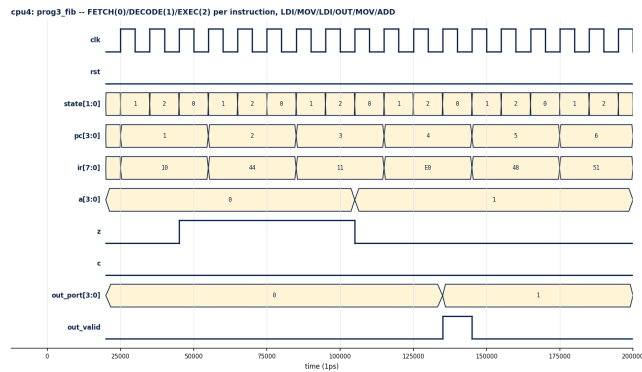
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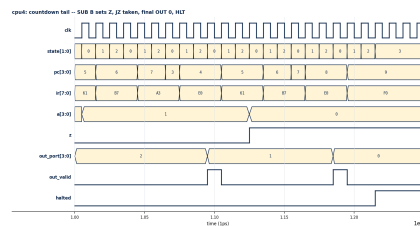
prog3_fib, first six instructions: state 0/1/2 = FETCH/DECODE/EXEC, IR 10 44 11 E0 48 51, A becomes 1 after LDI 1, OUT pulses out_valid with out_port = 1.

Synthesis and implementation results

The program ROM is constant-folded, so the flip-flops are PC, IR, state, four registers, two flags, the 64-bit RAM and the registered control word. The iCE40 critical path (from nextpnr) is the RAM read mux into the accumulator through the ALU; with CPI = 3 the processor executes 33.3 MIPS at 100 MHz.

Flow (all run here)	Measured result
Yosys 0.68 synth_ice40	362 cells: 222 SB_LUT4, 124 FF (72 DFFE, 40 DFFESR, 7 DFF, 5 DFFSR), 16 SB_CARRY
nextpnr-ice40 hx8k ct256, --freq 100 --seed 1	317 / 7680 LC (4 %), 32 SB_IO, 1 SB_GB; Fmax 115.86 MHz, PASS at 100 MHz
icetime -d hx8k	total path delay 9.06 ns (110.42 MHz)
Yosys synth_xilinx xc7 (stat)	204 cells: 97 LUT2-LUT6, 60 FDRE, 5 CARRY4, 4 RAM32M
Yosys + ABC, tools/cmos_cells.lib	826 cells, chip area 6205 um ² (126 DFF = 2520, 323 NAND, 187 NOR, 89 MUX2, 50 AND, 25 OR, 19 NOT, 7 XOR/XNOR)
tools/mini_sta.py (CMOS netlist)	critical path 2.35 ns, 18 logic levels, ~425 MHz; leakage 0.86 uW (teaching-grade model)

Control-flow waveforms



prog2_countdown tail: SUB B drives A to 0 and Z high in EXEC; the following JZ samples Z in DECODE and loads PC <= 7; the final OUT emits 0 and HLT parks the FSM in state 3 with halted = 1.

Dynamic instruction mix (from isa_sim trace)

Program	Instructions executed	Mix (dynamic count)	Branches taken / total	Time at 100 MHz
prog1_add	13	LDI 2, STA 2, LDA 2, MOV 1, ADD 2, IN 1, OUT 2, HLT 1	0 / 0	0.39 us
prog2_countdown	40	OUT 10, SUB 9, JZ 9, JMP 8, LDI 2, MOV 1, HLT 1	9 / 17 (JZ 1/9, JMP 8/8)	1.20 us
prog3_fib	44	MOV 14, OUT 7, ADD 7, JC 7, JMP 6, LDI 2, HLT 1	7 / 13 (JC 1/7, JMP 6/6)	1.32 us

Declared substitutions / not run here

- Vivado not available: synth/vivado_cpu4.tcl and synth/cpu4_basys3.xdc (Basys-3, xc7a35t, 100 MHz clock, switches as in_port, LEDs as out_port/flags) are provided and were not run. The Xilinx numbers above are Yosys synth_xilinx stat only.
- No FPGA board was programmed; synth/cpu4.asc is a real nextpnr/icepack-ready bitstream for the iCE40-HX8K breakout, not tested on silicon.
- CMOS figures use the shared educational liberty (180 nm-class representative values), not a foundry PDK. Simulation used Icarus Verilog, not ModelSim.

Learnings / next steps

Registering the control word in DECODE kept EXECUTE flat and the critical path in the datapath, not the decoder. The golden-model comparison forces assembler, ISA model and RTL to agree on operand packing (MOV). Next: a WRITEBACK state for synchronous block RAM, CALL/RET with a small stack, and a UART loader so programs are not baked into the bitstream.